Plane-Wave Image Reconstruction in Fourier Domain Using IQ Channel Data

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***Abstract*—Fourier-domain beamforming (FDB) can reconstruct plane-wave ultrasound images at a much lower computational cost compared to the conventional delay-and-sum (DAS) approach. The latter can handle both raw radio-frequency (RF) and in-phase/quadrature (IQ) channel data, while the former is typically formulated for RF input. As IQ beamforming is often preferable (e.g., baseband data processing), we propose a simple extension of RF-based FDB to handle IQ signals. Utilizing the PICMUS experimental dataset, we compare an IQ-based FDB implementation against its RF-based counterpart, by evaluating the reconstructed B-mode images using the generalized contrast-to-noise ratio and full width at half maximum metrics, which demonstrates virtually identical image quality in both cases.**

***Index Terms*—Plane-wave ultrasound imaging, Fourier-domain beamforming, IQ channel data.**

I. Introduction

Plane-wave ultrasound imaging enables rapid acquisition of channel-data frames using a limited number of steered plane wave (PW) emissions, coupled with coherent compounding of beamformed data across multiple steering angles to improve the resulting image quality [1]. Fast Fourier-domain beamforming (FDB) is well-suited for such imaging, wherein each channel-data frame is transformed and remapped using the frequency-wavenumber data representation (e.g., see [2]), as opposed to pixel-by-pixel reconstruction using conventional delay-and-sum (DAS) beamforming in the original spatio-temporal domain [3]. Here, we focus on Temme-Müller (TM) migration [4] as a representative example of FDB.

TM migration handles real-valued radio-frequency (RF) channel data, which allows for processing only the positive-frequency half-spectrum prior to analytic image formation [5]. This work addresses scenarios where the channel data consist of in-phase/quadrature (IQ) samples. Although IQ signals are complex-valued, they are typically sampled at a much lower rate following demodulation and low-pass filtering [6]. Consequently, the IQ representation minimizes data storage and transfer bandwidth requirements while fully preserving the complex phase information critical for coherent reconstruction.

Directly applying RF-based TM migration to IQ channel data yields incorrect results unless the frequency shift induced by demodulation is properly resolved. While TM remapping equations involve the physical (temporal) frequency and aperture wavenumber, only the former is affected by IQ demodulation: its baseband frequencies f_b correspond to the physical frequencies $F = f_b + f_d$, where f_d denotes the demodulation frequency. Therefore, the RF-based TM equations can easily be adapted to IQ data by correcting frequency bins to F when applying PW angle compensation and spectral remapping.

This paper presents a formulation of IQ-based TM migration for PW imaging that incorporates the demodulation frequency into the frequency-wavenumber data processing. The proposed method is evaluated on *in-vitro* datasets from PICMUS [7] using two imaging cases: a single (zero-angle) PW emission, and 75 PW emissions at steering angles ranging from -16° to $+16^\circ$. Our IQ-based TM method is compared with its RF-based counterpart as well as a conventional DAS beamformer operating on IQ channel data (see Section III).

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II. Fourier-Domain Beamforming

For each plane-wave emission steered at angle θ, we assume that a linear transducer array records a complex-valued 2D IQ channel-data frame $D_\theta^{\mathrm{IQ}}(t,x)$, where t refers to temporal sampling instances and x denotes receive-element positions along the aperture [1]. The size of D_θ^{IQ} is $N_t \times N_x$, where N_t is the number of temporal samples, and N_x is the number of receive elements.

Let $D_\theta^{\mathrm{RF}}(t,x)$ represent a real-valued 2D RF channel-data frame, with its "analytic" representation denoted by $D_\theta^{+}(t,x)$. IQ demodulation translates the spectrum by f_d, followed by low-pass filtering (LPF) [6]:

$$D_\theta^{\mathrm{IQ}}(t,x) = \mathrm{LPF}\left\{D_\theta^{+}(t,x)\exp(-j2\pi f_d t)\right\}, \tag{1}$$

so that

$$D_\theta^{+}(t,x) = D_\theta^{\mathrm{IQ}}(t,x)\exp(j2\pi f_d t), \tag{2}$$

$$D_\theta^{\mathrm{RF}}(t,x) = \mathrm{Re}\{D_\theta^{+}(t,x)\}. \tag{3}$$

Consequently, the relationship between the corresponding RF spectrum S_θ^{RF} and baseband IQ spectrum S_θ^{IQ} is as follows:

$$S_\theta^{\mathrm{IQ}}(f_b,k_x) = S_\theta^{\mathrm{RF}}(f_b+f_d,k_x). \tag{4}$$

The proposed migration method uses this simple relationship to adjust RF-based TM equations to IQ data processing.

A. RF-Based TM Migration

The original TM migration method [4] can be viewed as a simplified version of well-known Lu's beamforming [2]. For any given PW angle θ, it processes the entire RF channel-data frame $D_\theta^{\mathrm{RF}}(t,x)$ at once, as outlined below.

1) Compute the 1D Fourier transform of $D_\theta^{\mathrm{RF}}(t,x)$ along t to produce intermediate spectrum $S_\theta^{\mathrm{RF}}(f,x)$. Next, compute its angle-corrected version:

$$S_\theta^{\mathrm{RF}}(f,x) \leftarrow S_\theta^{\mathrm{RF}}(f,x)\cdot\exp(j2\pi f x\sin(\theta)/c), \tag{5}$$

where c denotes the speed of sound. Then, compute the 1D Fourier transform of $S_\theta^{\mathrm{RF}}(f,x)$ along x, which yields pre-migration frequency-wavenumber spectrum $S_\theta^{\mathrm{RF}}(f,k_x)$.

2) Remap $S_\theta^{\mathrm{RF}}(f,k_x)$ onto the (k_z,k_x) grid coordinates to get migrated spectrum $\widetilde{S}_\theta^{\mathrm{RF}}(k_z,k_x)$:

$$\widetilde{S}_\theta^{\mathrm{RF}}(k_z,k_x) = A_\theta(k_z,k_x)S_\theta^{\mathrm{RF}}(\widetilde{f}_\theta(k_z,k_x),k_x), \tag{6}$$

where the required frequency $\widetilde{f}_\theta(k_z,k_x)$ and scaling factor $A_\theta(k_z,k_x)$ are given by

$$\widetilde{f}_\theta(k_z,k_x) = \frac{ck_z\left[1+(k_x/k_z)^2\right]}{2\left[\cos(\theta)+\sin(\theta)k_x/k_z\right]}, \tag{7}$$

$$A_\theta(k_z,k_x) = \frac{c\left|\cos(\theta)\left[1-(k_x/k_z)^2\right]+2\sin(\theta)k_x/k_z\right|}{2\left[\cos(\theta)+\sin(\theta)k_x/k_z\right]^2}. \tag{8}$$

3) Sum the migrated spectra over all PW angles to obtain the coherently compounded image spectrum:

$$\widetilde{S}_{\mathrm{cc}}^{\mathrm{RF}}(k_z,k_x) = \sum_\theta \widetilde{S}_\theta^{\mathrm{RF}}(k_z,k_x). \tag{9}$$

4) Compute the 2D inverse Fourier transform of $\widetilde{S}_{\mathrm{cc}}^{\mathrm{RF}}(k_z,k_x)$, which yields final image $D_{\mathrm{cc}}(z,x)$ with $z=ct/2$.

B. IQ-Based TM Migration

Since RF channel data are real-valued, their negative temporal frequencies are redundant. Therefore, only the positive-frequency half-spectrum is required to perform RF-based TM migration. In contrast, IQ channel data are complex-valued baseband signals and do not have conjugate symmetry. Consequently, our IQ-based TM migration uses the full two-sided baseband spectrum and maps it to physical frequencies through $F=f_b+f_d$. By re-indexing baseband frequency bins to shift from f_b to F, we can rewrite (5) and (6) as follows:

$$\widetilde{S}_\theta^{\mathrm{IQ}}(F,x) \leftarrow \widetilde{S}_\theta^{\mathrm{IQ}}(F,x)\exp(j2\pi F x\sin(\theta)/c)\,. \tag{10}$$

$$\widetilde{S}_\theta^{\mathrm{IQ}}(K_z,k_x) = A_\theta(K_z,k_x)\widetilde{S}_\theta^{\mathrm{IQ}}(\widetilde{f}_\theta(K_z,k_x),k_x), \tag{11}$$

where $K_z = 2F/c$. Equations (7) and (8) remain the same by substituting k_z with K_z. The final coherently compounded image is given by

$$D_{\mathrm{cc}}^{\mathrm{IQ}}(z,x) = \mathcal{F}_{\mathrm{2D}}^{-1}\left\{\sum_\theta \widetilde{S}_\theta^{\mathrm{IQ}}(K_z,k_x)\right\}, \tag{12}$$

where $\mathcal{F}_{\mathrm{2D}}^{-1}$ represents the 2D inverse Fourier transformation.

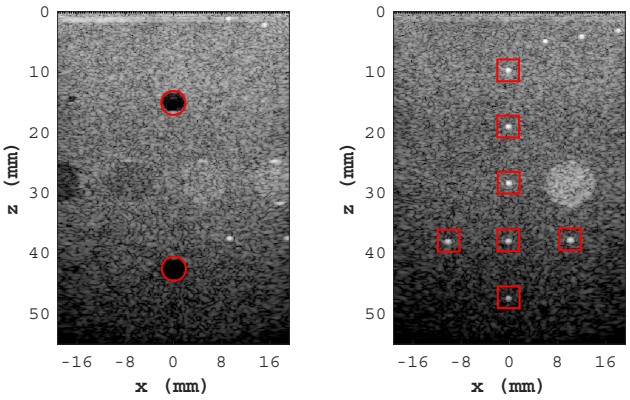


Fig. 1. 75-PW coherently compounded image examples with marked evaluation regions: TYPE-1 (left) and TYPE-2 (right), 60-dB range. These B-mode images have been reconstructed from IQ channel data using a basic PICMUS-provided DAS beamformer (1.75 f-number, Tukey-25% apodization) [7].

III. EVALUATION RESULTS

We evaluated our IQ-based TM migration method on two PICMUS *in-vitro* datasets [7], labeled TYPE-1 and TYPE-2. The former contains two anechoic targets used for contrast assessment, while the latter contains seven point-like targets used for resolution assessment. They are illustrated in Fig. 1.

Each frame of RF channel data has $N_t\times N_x = 1536\times 128$ samples. In contrast, each frame of IQ channel data has $(384\times 128)\cdot 2$ samples, representing real and imaginary components of IQ signals that have been sampled at a lower rate. Thus, the total number of IQ data samples is 50% smaller than in the RF case.

For RF data, the positive-frequency half-spectrum contained 2048 frequency bins. For IQ data, the full two-sided baseband spectrum contained 1024 frequency bins. In other words, the Fourier grid used during RF-based TM migration was twice as large as its IQ counterpart.

Three reconstruction methods were compared: a basic DAS beamformer using IQ channel data (DAS-IQ), TM migration using RF channel data (TM-RF), and TM migration using IQ channel data (TM-IQ). Two imaging cases were considered: a single zero-angle PW emission, and 75 PW emissions steered at equispaced angles θ ranging from -16° to $+16^\circ$. For TYPE-1 images, we measured the generalized contrast-to-noise ratio (gCNR) values for both top and bottom targets. For TYPE-2 images, we measured the mean axial and lateral full width at half maximum (FWHM) values over all seven targets. These measurements are listed in Table I, and Fig. 2–5 display the corresponding TM-reconstructed images.

In the single-PW case, TM-IQ yields gCNR values of 0.876 and 0.830 for the top and bottom TYPE-1 targets, virtually matching TM-RF's 0.875 and 0.827, respectively. Similarly, the axial and lateral FWHM values of 0.482 and 0.788 mm given by TM-IQ are nearly the same as TM-RF's 0.479 and 0.793 mm, respectively. For DAS-IQ, the gCNR and lateral FWHM values are noticeably worse compared to the Fourier-domain reconstruction results.

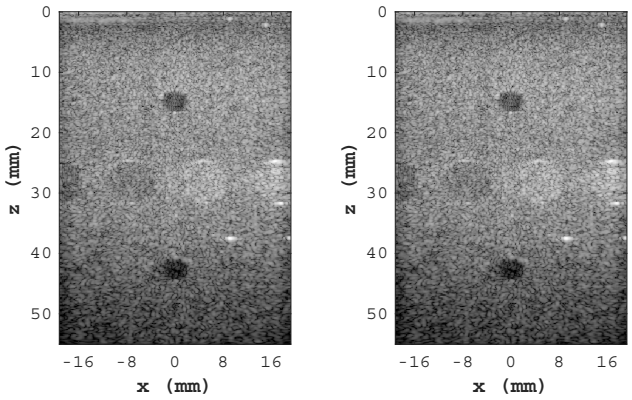


Fig. 2. TYPE-1 images, single PW: TM-RF (left) and TM-IQ (right).

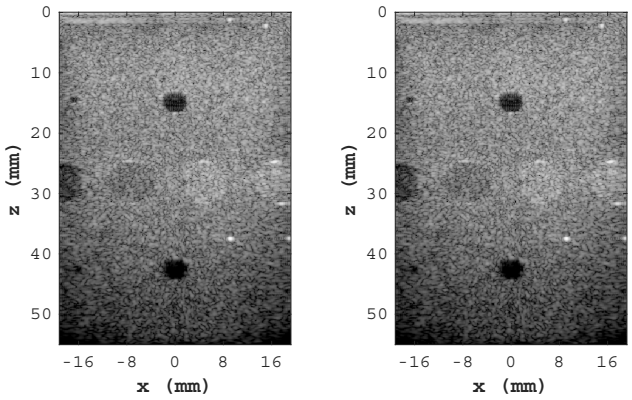


Fig. 3. TYPE-1 images, 75 PWs: TM-RF (left) and TM-IQ (right).

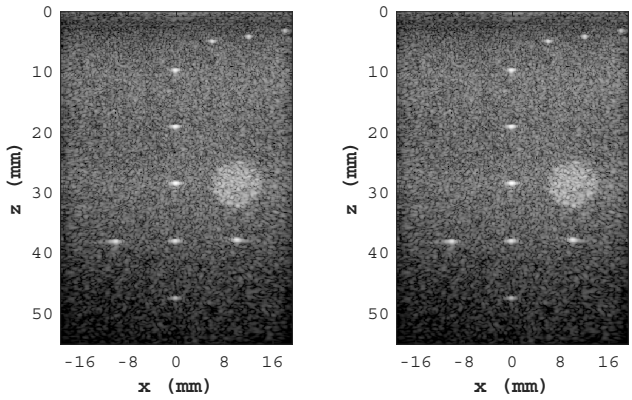


Fig. 4. TYPE-2 images, single PW: TM-RF (left) and TM-IQ (right).

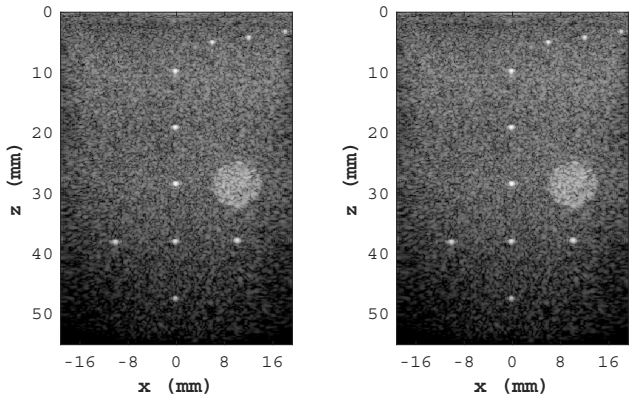


Fig. 5. TYPE-2 images, 75 PWs: TM-RF (left) and TM-IQ (right).

TABLE I
IMAGE QUALITY ASSESSMENT FOR DAS-IQ, TM-IQ, AND TM-RF RECONSTRUCTION METHODS.

Channel Data Encoding		gCNR		$\overline{\text{FWHM}}$, mm	
		Top	Bottom	Axial	Lateral
1 PW	DAS-IQ	0.846	0.813	0.482	0.953
	TM-RF	0.875	0.827	0.479	0.793
	TM-IQ	0.876	0.830	0.482	0.788
75 PWs	DAS-IQ	0.971	0.951	0.483	0.613
	TM-RF	0.971	0.962	0.482	0.528
	TM-IQ	0.968	0.955	0.486	0.527

In the 75-PW case, TM-IQ yields the top- and bottom-target gCNR values of 0.968 and 0.955, respectively, versus 0.971 and 0.962 produced by TM-RF. The respective axial and lateral FWHM values are 0.486 and 0.527 mm versus 0.482 and 0.528 mm. This observation confirms that the reconstruction quality differences between TM-IQ and TM-RF are negligible in the multiangle acquisition scenario as well. Relative to DAS-IQ, both TM-RF and TM-IQ provide better lateral FWHM with similar axial FWHM and gCNR. It is worth highlighting that coherent compounding significantly improves contrast and lateral resolution for all three methods.

IV. CONCLUSION

We have presented an IQ-based formulation of TM migration for Fourier-domain plane-wave image reconstruction. It operates directly on demodulated in-phase/quadrature channel data. Evaluation utilizing two PICMUS datasets demonstrate that this method yields image quality virtually identical to its original RF-based counterpart. The proposed approach allows FDB to benefit from the reduced sampling rate of baseband IQ data, thereby reducing both channel-data storage requirements and Fourier grid dimensions.

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